Feature Engineering & Feature Selection Validation Report

Model Name: SMI3.0 Mortgage Insurance Credit Risk Model

1. Feature Creation and Transformation

Objective

To assess whether the engineered features are economically meaningful, appropriately transformed, and suitable for mortgage insurance credit risk modelling.

Description

Technical Model Validation reviewed the feature engineering framework to assess whether derived variables, borrower-level metrics, ratios, and transformations appropriately represent mortgage credit risk while preserving the economic meaning of the original data.

Validation Assessment

Review Item	**Status**
Engineered features documented	Partial
Business rationale appropriate	Yes
Transformations preserve economic meaning	Yes
Derived variables appropriate	Yes
Features aligned with intended use	Yes
No material distortion of risk relationships	Yes

Validator Comments

The feature engineering framework creates variables representing borrower affordability, property characteristics, credit quality, and regional economic conditions.

The engineered variables are consistent with mortgage underwriting practices and support the intended modelling objective. The applied transformations preserve the economic interpretation of the underlying variables and do not introduce unnecessary complexity.

Additional documentation describing feature definitions and business rationale would further strengthen governance.

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2. Missing Value Related Features

Objective

To assess whether missing-value treatment and related feature construction are appropriate and consistently applied.

Description

Technical Model Validation reviewed the missing-value methodology to assess whether missing information is handled consistently and whether the applied treatment could introduce bias or affect feature behaviour.

Validation Assessment

Review Item	Status
Missing-value indicators considered	Partial
Missingness assessed as informative	Partial
Imputation methodology documented	Partial
No future information used	Yes
Consistent treatment across datasets	Yes

Validator Comments

The framework applies structured fallback methods for important variables, including borrower credit scores and property-related information.

The review did not identify any evidence that future information is used during missing-value treatment. The overall approach is considered appropriate for the intended modelling purpose.

Documentation describing the missing-value methodology and supporting rationale should be expanded to improve transparency and governance.

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3. Outlier-Related Features and Handling

Objective

To assess whether extreme observations are handled appropriately without affecting meaningful mortgage credit risk information.

Description

Technical Model Validation reviewed the treatment of extreme values during feature engineering to determine whether the methodology is consistent and preserves valid risk information.

Validation Assessment

Review Item	Status
Extreme values reviewed	Yes
Capping methodology documented	Partial
Tail-risk observations preserved	Partial
Outlier indicators justified	Partial
Consistent treatment across datasets	Yes

Validator Comments

The framework applies defined limits to selected variables, including credit scores, property values, contract rates, and loan terms.

The overall approach is considered reasonable and consistent with mortgage underwriting practices. However, additional documentation supporting selected thresholds and clipping rules would improve governance.

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4. Categorical Variable Handling

Objective

To assess whether categorical variables are grouped and standardized appropriately for model development.

Description

Technical Model Validation reviewed the handling of categorical variables to determine whether category definitions are consistent, stable, and suitable for mortgage credit risk modelling.

Validation Assessment

Review Item	Status
Category grouping documented	Partial
Rare categories handled appropriately	Partial
Variable treatment appropriate	Yes
Encoding suitable for model	Yes
Encoding avoids leakage	Yes
Category definitions stable	Partial

Validator Comments

The framework standardizes several business categories, including occupancy, loan purpose, portability, tenure, regional assignment, and rural classification.

Unknown values are handled using consistent business rules. The overall approach is appropriate for the intended modelling objective.

Documentation of category grouping rules and business rationale should be strengthened to support future governance and validation activities.

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5. Time-Based Feature Engineering

Objective

To assess whether time-based features are constructed appropriately and do not introduce future information.

Description

Technical Model Validation reviewed the construction of date-based and economic variables to determine whether only information available at the underwriting date is used during feature generation.

Validation Assessment

Review Item	Status
Date variables handled appropriately	Yes

Review Item	Status
Lag features use prior information	Partial
Rolling features appropriate	Partial
Vintage effects considered	Yes
No future information introduced	Partial

Validator Comments

The framework incorporates application timing, portfolio vintage, and regional economic information into feature generation.

No material conceptual concerns were identified during the review. However, additional validation should confirm that forward-looking economic variables are aligned with the underwriting date and do not introduce unintended data leakage.

This completes **Sections 1–5**. The next part will include:

- **6. Interaction and Ratio Features**
- **7. Binning and Discretization**
- **8. Feature Leakage Assessment**
- **9. Feature Redundancy and Correlation**
- **10. Feature Selection Methodology**
- **11. Stability and Robustness**
- **12. Interpretability and Governance**
- **13. Overall Conclusion, Findings, and Required Actions**

This will complete the report in the same style as the previous SMI2.0 validation report.

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Great. Below is the **second half** of the **Feature Engineering & Feature Selection Validation Report**, written in the same concise validation style as the previous SMI2.0 Technical Validation Report.

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6. Interaction and Ratio Features

Objective

To assess whether interaction and ratio features are mathematically correct, economically meaningful, and appropriate for mortgage insurance credit risk modelling.

Description

Technical Model Validation reviewed the affordability ratios, borrower-level combinations, and interaction features to determine whether they appropriately represent borrower risk and improve model performance without adding unnecessary complexity.

Validation Assessment

Review Item	Status
Interaction terms meaningful	Yes
Ratio calculations appropriate	Yes
Multicollinearity assessed	Partial
Interaction features improve interpretation	Yes
Number of interaction features appropriate	Yes

Validator Comments

The feature engineering framework includes affordability ratios, borrower-level aggregation measures, property-related variables, and weighted credit metrics that are consistent with mortgage underwriting practices.

The interaction and ratio features are economically meaningful and support the intended modelling objective. No excessive feature complexity was identified during the review.

Additional correlation analysis would provide further assurance that highly correlated features do not affect model stability.

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7. Binning and Discretization

Objective

To assess whether grouped variables and discretized features are stable, interpretable, and supported by business rationale.

Description

Technical Model Validation reviewed the grouping and discretization methods used during feature engineering to determine whether they remain consistent and interpretable over time.

Validation Assessment

Review Item	Status
Bin definitions documented	Partial
Business rationale appropriate	Yes
Bins interpretable	Yes
Bin boundaries reasonable	Yes
Consistent across datasets	Yes

Validator Comments

The framework relies primarily on business-driven grouping and normalization rather than extensive statistical binning.

Variables such as loan purpose, tenure, and geographic classification are grouped into business-relevant categories that support model interpretation.

The overall approach is considered appropriate. Additional documentation of grouping rules would improve governance.

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8. Feature Leakage Assessment

Objective

To assess whether engineered features contain direct or indirect future information that could artificially improve model performance.

Description

Technical Model Validation reviewed the feature engineering process to determine whether the selected variables respect the underwriting decision date and avoid target leakage.

Validation Assessment

Review Item	Status
Candidate features reviewed for leakage	Partial
No post-outcome information identified	Yes
No proxy outcome variables identified	Partial
Features respect decision date	Partial
Formal leakage testing completed	Partial

Validator Comments

The review did not identify any obvious evidence of target leakage within the feature engineering framework.

However, the use of forward-looking economic variables requires additional validation to confirm that all features are aligned with the underwriting date and do not unintentionally introduce future information.

Formal leakage testing should be completed before final model approval.

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9. Feature Redundancy and Correlation

Objective

To assess whether redundant or highly correlated variables could reduce model stability or interpretability.

Description

Technical Model Validation reviewed the feature set to determine whether overlapping information or excessive feature complexity could affect model robustness.

Validation Assessment

Review Item	Status
Highly correlated variables identified	Partial
Redundant features reviewed	Partial
Correlation impact assessed	Partial

Review Item	Status
Overlapping information justified	Partial
Overall feature complexity appropriate	Yes

Validator Comments

The final feature set combines borrower, property, credit, and economic information in a structured manner.

No material concerns relating to excessive feature complexity were identified. However, additional correlation analysis should be performed to identify redundant variables and improve model interpretability.

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10. Feature Selection Methodology

Objective

To assess whether the feature selection process is appropriate, explainable, and supported by business and statistical evidence.

Description

Technical Model Validation reviewed the feature selection methodology to determine whether the final feature set supports the intended modelling objective and remains operationally practical.

Validation Assessment

Review Item	Status
Feature selection criteria documented	Partial
Selection methodology appropriate	Yes
Business knowledge considered	Yes
Statistical selection consistently applied	Partial
Final feature set appropriate	Yes
Excluded features documented	No

Validator Comments

The selected features represent borrower affordability, credit quality, property characteristics, regional economic conditions, and policy information.

The overall feature selection approach is appropriate for mortgage insurance credit risk modelling.

Formal documentation describing feature selection criteria and excluded variables would strengthen governance and improve future model maintenance.

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11. Feature Stability and Robustness

Objective

To assess whether engineered features remain stable across different time periods and portfolio segments.

Description

Technical Model Validation reviewed the feature engineering framework to determine whether the selected variables are expected to remain stable over time and under changing portfolio conditions.

Validation Assessment

Review Item	Status
Feature distributions reviewed over time	Partial
Stability assessed across segments	Partial
Features stable across development periods	Partial
Features robust to small data changes	Yes
Stability concerns documented	Partial

Validator Comments

The framework incorporates multiple portfolio years and regional segmentation, providing a foundation for future stability analysis.

However, formal feature stability monitoring and population stability testing were not identified during the current review.

Additional monitoring procedures are recommended to support ongoing model governance.

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12. Interpretability and Governance

Objective

To assess whether the engineered features are explainable, governable, and suitable for ongoing monitoring.

Description

Technical Model Validation reviewed whether the final feature set can be understood by business, validation, and governance stakeholders and whether it supports long-term model monitoring.

Validation Assessment

Review Item	Status
Features understandable to stakeholders	Yes
Features comply with governance standards	Partial
Features appropriate for model materiality	Yes
Expert judgement documented	Partial
Features support ongoing monitoring	Yes

Validator Comments

Most engineered features are economically intuitive and directly related to mortgage underwriting.

The overall feature framework supports model interpretation and ongoing validation activities.

Further documentation of business rules and feature governance would improve transparency and facilitate future model reviews.

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13. Overall Conclusion

Validation Assessment

Conclusion Item	Status
Feature engineering acceptable	Yes
Feature engineering acceptable subject to remediation	Yes
Feature engineering not acceptable	No

Conclusion

Technical Model Validation reviewed the feature engineering and feature selection framework supporting the SMI3.0 Mortgage Insurance Credit Risk Model.

The review confirmed that the engineered features are generally appropriate for mortgage insurance credit risk modelling. The feature engineering process incorporates borrower, property, credit bureau, and economic information in a structured and reproducible manner. The selected features are economically meaningful, support the intended modelling objective, and are consistent with mortgage underwriting practices.

No material issues were identified with the overall feature engineering approach.

Several opportunities for improvement were identified, including enhanced documentation of feature engineering decisions, formal feature selection methodology, feature stability monitoring, correlation analysis, and additional validation of potential data leakage associated with time-dependent economic variables.

Overall, Technical Model Validation concludes that the feature engineering and feature selection framework is fit for use, subject to completion of the recommended governance and validation improvements.

Key Findings

- The feature engineering framework is consistent with mortgage underwriting principles.
- Engineered features preserve the economic meaning of the underlying data.
- The selected feature set is appropriate for the intended modelling objective.
- No material feature engineering concerns were identified during the review.

- Additional documentation, correlation analysis, feature stability assessment, and leakage testing would strengthen model governance.

Required Actions

Action	Priority	Owner
Document feature engineering and selection methodology	Moderate	Model Development
Complete formal feature leakage assessment	High	Validation Team
Perform feature correlation analysis	Moderate	Model Development
Document feature transformation rationale	Moderate	Model Development
Implement feature stability monitoring	Moderate	Model Risk
Validate timing of economic variables	High	Model Development